

Zero-Trust Blockchain-Enabled Secure Next-Generation Healthcare Communication Network

Hai Zhu, Xingsi Xue^{ID}, *Member, IEEE*, Mengmeng Xu, Byung-Gyu Kim^{ID}, *Senior Member, IEEE*, Xiaohong Lyu^{ID}, and Shalli Rani^{ID}

Abstract—Conventional security architectures and models are considered single-network architecture solutions, which assume that devices authenticated within the network are implicitly trusted. However, such an approach is unsuitable for next-generation networks (NGNs). Zero-trust security was introduced to overcome these challenges using context-aware, dynamic, and intelligent authentication schemes. This paper proposes a novel zero-trust blockchain-enabled framework for secure next-generation healthcare communication network (HCN). The proposed framework integrates zero-trust and blockchain to provide a decentralized, secure, and intelligent solution for healthcare communication in NGNs. The system model comprises three components: HCN user identity modeling, blockchain and risk assessment-based access control, and dynamic trust gateway. The user identity modeling component utilizes attribute-based user behavior trajectory features, while the access control component leverages smart contracts-based risk assessment. The dynamic trust gateway component employs a consensus mechanism to achieve dynamic gateway switching and enhance network resilience. Simulation results demonstrate that the proposed framework achieves 31% lower calculation delays, 3% higher trust values, and 3% better attack detection accuracy compared to best baseline methods. It also exhibits a 2% improvement in access control granularity and maintains 95% network throughput under various failure scenarios.

Index Terms—Zero-trust security, healthcare communication network, next-generation networking, blockchain.

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Hai Zhu and Mengmeng Xu are with the School of Computer, Henan University of Engineering, Zhengzhou 451191, China (e-mail: zhuhai@xjtu.edu.cn; mmxu873@163.com).

Xingsi Xue is with the School of Computer Science and Mathematics, Fujian University of Technology, Fuzhou 350118, China (e-mail: jack8375@gmail.com).

Byung-Gyu Kim is with the Department of AI Engineering, Sookmyung Women's University, Seoul 04310, Republic of Korea (e-mail: bg.kim@sookmyung.ac.kr).

Xiaohong Lyu is with the Department of Imaging, The First Affiliated Hospital of Jinzhou Medical University, Jinzhou 121002, China (e-mail: lvxh@jzmu.edu.cn).

Shalli Rani is with the Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab 140401, India (e-mail: shalli.rani@chitkara.edu.in).

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I. INTRODUCTION

THE RAPID advancement of next-generation networking (NGN) technologies, including 5G, 6G, and beyond, is revolutionizing the healthcare industry by enabling intelligent and connected healthcare services [1], [2]. Next-generation healthcare communication network (HCN) refers to an advanced, interconnected system that leverages emerging technologies such as 5G, Internet of Things (IoT), artificial intelligence, and blockchain to facilitate seamless, secure, and efficient communication and data exchange in healthcare settings. Next-generation HCN aims to enable real-time health monitoring, telemedicine, personalized care, and data-driven decision-making while ensuring high levels of security and privacy. NGN-enabled HCN offers ubiquitous connectivity, high bandwidth, and low latency, which facilitate the implementation of various innovative healthcare applications, such as telemedicine, remote monitoring, and personalized medicine [3], [4], [5]. These applications can improve patient care, enhance healthcare accessibility, and optimize resource utilization. In this paper, the NGNs in healthcare are defined as advanced communication infrastructures that integrate 5G/6G technologies, IoT devices, and cloud computing to enable high-speed, low-latency, and secure data transmission for various healthcare applications such as telemedicine, remote patient monitoring, and AI-assisted diagnostics. However, the increasing complexity and heterogeneity of NGN-enabled HCN also introduce new security challenges and vulnerabilities that must be addressed to ensure the confidentiality, integrity, and availability of sensitive healthcare data. The diverse range of connected devices, the vast amount of data generated, and the critical nature of healthcare services make these networks attractive targets for cybercriminals, necessitating the development of robust and adaptive security measures [6].

Conventional security architectures and models often rely on a single network architecture solution, assuming that devices authenticated within the network can be implicitly trusted [7], [8]. These traditional approaches presume that once a device has been authenticated, it is free to access, move, or exfiltrate data without further scrutiny. However, this paradigm may introduce significant security threats and attacks, as it fails to account for the possibility of compromised devices or malicious insiders. While such an approach may be

suitable for certain network scenarios with limited complexity and a well-defined perimeter, it needs to be equipped to handle the challenges posed by NGNs. To address these shortcomings, zero-trust security has emerged, which abandons the reliance on entry-point authentication in favor of context-aware, dynamic, and intelligent authentication schemes that continuously monitor and assess the risk associated with each device and user interaction [9], [10], [11], [12]. Zero-trust is a security concept that assumes no user, device, or network should be automatically trusted, regardless of location or previous authentication status. It requires continuous verification of every access attempt and enforces least-privilege access principles.

Despite the promising potential of zero-trust security in securing NGNs, research in this area is still in its early stages, with limited work exploring the application of zero-trust models to NGNs. To fully realize the benefits of zero-trust security in NGNs, two key elements must be considered: threat intelligence and decentralized authentication [13], [14]. Threat intelligence provides valuable insights into potential security risks and enables proactive defense mechanisms, while decentralized authentication eliminates the reliance on a single point of failure and enhances the overall security of the system. Additionally, continuous and dynamic trust evaluation is essential to achieve high levels of access control in NGNs, ensuring that access privileges are granted based on real-time assessment of device and user trustworthiness. Furthermore, the adoption of decentralized authentication methods, such as blockchain technology, can provide secure and tamper-proof storage and data sharing, further bolstering the security of NGNs [15], [16], [17].

To address the above challenges, this paper proposes a novel zero-trust blockchain-enabled framework for secure next-generation HCN. The main contributions of this paper are summarized as follows:

1. We propose a novel zero-trust blockchain-enabled framework for secure next-generation HCN (ZTB-NGHCN). The proposed framework integrates zero-trust and blockchain to provide a decentralized, secure, and intelligent solution for healthcare communication in NGNs.
2. We design a HCN user identity modeling component that utilizes attribute-based user behavior trajectory features to establish user identities. The user identity modeling considers various attributes and employs a sliding window mechanism and punishment mechanism to enhance security.
3. We develop a blockchain and risk assessment-based access control component that leverages smart contracts risk assessment to provide decentralized and intelligent access control. The access control component incorporates a dynamic permission threshold adjustment algorithm to adapt to changing network conditions.
4. We introduce a dynamic trust gateway component that employs a consensus mechanism to achieve dynamic gateway switching and enhance network resilience. The dynamic trust gateway can effectively resist common attacks in HCN.

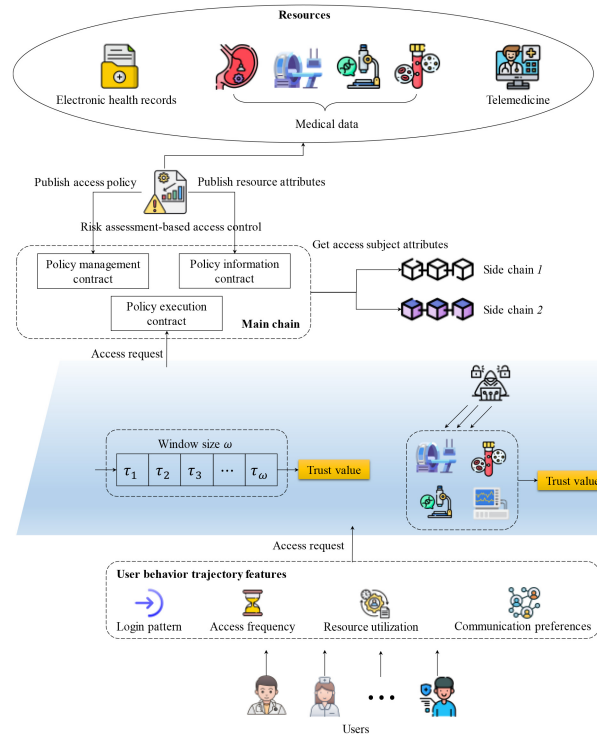


Fig. 1. System model of ZTB-NGHCN.

The remainder of this paper is organized as follows. Section II presents the system model of the proposed framework. Section III describes the zero-trust next-generation HCN. Section IV provides the simulation results and analysis. Finally, Section V concludes the paper.

II. SYSTEM MODEL

This section presents the system model of the proposed ZTB-NGHCN framework. The system model follows the analysis of the zero-trust network security scenarios and the design of the zero-trust network security architecture in healthcare networks [12], [13], [18]. The system model consists of three main components: HCN user identity modeling, blockchain and risk assessment-based access control, and dynamic trust gateway, as shown in Fig. 1.

The ZTB-NGHCN framework is designed to address a comprehensive threat model that includes 1) External attackers attempting to gain unauthorized access, 2) Insider threats from compromised devices or malicious users, 3) Man-in-the-middle attacks on network communications, and 4) Distributed denial of service attacks targeting critical healthcare services.

The ZTB-NGHCN framework is particularly beneficial for healthcare use cases involving sensitive data exchange: 1) Multi-institutional cancer research collaborations, 2) Nationwide electronic health record systems, and 3) Telemedicine platforms for rural healthcare delivery. In these scenarios, the framework's robust security and fine-grained access control capabilities ensure data protection while facilitating necessary information sharing.

While our framework is designed with current and near-future network technologies in mind, the advent of 6G is expected to enhance its capabilities further. 6G's ultra-low

latency and high bandwidth could enable real-time, AI-driven trust assessments and more sophisticated blockchain consensus mechanisms. Additionally, 6G's improved network slicing capabilities could allow for more granular and dynamic security policies within the ZTB-NGHCN framework.

A. HCN User Identity Modeling

The HCN user identity modeling component utilizes attribute-based user behavior trajectory features to establish user identities. The user identity modeling follows the zero-trust security architecture deployment, identity attribute definition, and user behavior trajectory feature extraction.

To accommodate the growing use of wearable health devices and consumer-grade health monitoring tools, the ZTB-NGHCN framework includes a consumer device integration layer (CDIL). The CDIL employs a combination of device fingerprinting and behavioral analysis to authenticate consumer devices. Data from these devices is validated through AI-driven plausibility checks before being integrated into the main health record. The blockchain component of the framework maintains a separate ledger for consumer device data, allowing for granular access control and easy auditing of this potentially less reliable data source.

To deploy the zero-trust security architecture in the HCN, we first select a base point G , whose order is n . Then, we randomly select an integer S as the private key and calculate the corresponding public key P .

$$P = S * G. \quad (1)$$

where $*$ denotes the multiplication operation on the elliptic curve. The public key P is managed by the certificate authority of the HCN and can be disclosed to the public, while the private key S is kept securely by the user.

In the context of NGN-enabled HCN, deploying a zero-trust security architecture is essential to ensure the security and privacy of sensitive healthcare data. By leveraging the properties of elliptic curve cryptography, the proposed architecture establishes a secure foundation for user identity management and authentication [19], [20], [21].

By integrating zero-trust principles with the NGN-enabled HCN, the proposed security architecture enables fine-grained access control, continuous authentication, and real-time monitoring of user activities. This comprehensive security framework ensures that only authorized users can access the required healthcare resources while maintaining the data's confidentiality, integrity, and availability.

In the zero-trust NGN-enabled HCN, the identity attribute refers to the unique original identification code assigned to each network user. Let μ_1 denote the original identification code of the healthcare network user, μ_σ denote the compiled user attribute code, and σ denote the probability of user information repetition in the HCN. The definition of the user identity attribute can be expressed as follows.

$$\Phi(\mu_1 \rightarrow \mu_\sigma) = \sum_{\sigma=1}^{\infty} \frac{\|\rho_\sigma \mu_\sigma - \rho_1 \mu_1\|^2}{\lambda_2^2 - \lambda_1^2}. \quad (2)$$

where ρ_1 denotes the original code compilation coefficient in the HCN, ρ_σ denotes the attribute code compilation coefficient, λ_1 denotes the original code matching parameter, and λ_2 denotes the attribute code matching parameter. This definition ensures that each user in the HCN has a unique and distinguishable identity attribute.

The probability of user information repetition, denoted by σ , is a critical factor in maintaining the uniqueness and integrity of user identity attributes within the HCN. By minimizing the likelihood of information repetition, the system can prevent identity spoofing and ensure that each user has a distinct and verifiable identity.

The compilation coefficients ρ_1 and ρ_σ transform the original identification code and the attribute code, respectively. These coefficients can be derived from a combination of user-specific information and system-generated parameters, enhancing the complexity and security of the identity attribute definition process.

The matching parameters λ_1 and λ_2 are employed to verify the consistency and authenticity of the original identification code and the compiled attribute code. By comparing these parameters, the system can detect any discrepancies or unauthorized modifications to the user identity attributes, thereby maintaining the integrity of the identity management system.

As expressed in eq. (2), the identity attribute definition provides a robust and flexible framework for managing user identities within the zero-trust NGN-enabled HCN. By incorporating healthcare-specific parameters and leveraging the strengths of the zero-trust security architecture, this approach ensures that user identities are accurately represented, securely managed, and effectively utilized for access control and authentication purposes.

User behavior trajectory features play a significant role in zero-trust security, as they can be used to determine the constraint capability of the authentication protocol on the identity attribute information in HCN. Let ξ denote the cross-domain transmission measure value of user identity information in the healthcare network, which can be calculated as follows.

$$\xi = \psi \left(\frac{H_1 + H_2 + \dots + H_n}{\bar{H}} \right)^2. \quad (3)$$

where $H_1, H_2, \dots, H_n$ denote n different user identity information samples in the HCN, $\bar{H}$ denotes the mean value of the information samples, and ψ denotes the information sample matching permission in the healthcare context.

Let ϑ denote the base parameter of user identity information value in the healthcare network, ν denote the user behavior parameter, and ζ denote the user behavior trajectory distribution parameter. The user behavior trajectory feature in the HCN can be expressed as follows.

$$\Omega = \Phi(\mu_1 \rightarrow \mu_\sigma) \times \xi(\nu + \zeta)^2 \cdot \|\Delta\eta\|. \quad (4)$$

where $\Delta\eta$ denotes the unit coding duration of the HCN user identity attribute information.

In HCN, user behavior trajectory features encompass various aspects, such as login patterns, access frequency, resource utilization, and communication preferences [22]. These features provide valuable insights into user behavior and help

identify deviations from expected patterns, which may indicate unauthorized access attempts or malicious activities.

The cross-domain transmission measure value, denoted by ξ , quantifies how user identity information is shared and transmitted across different domains within the HCN [23]. This measure takes into account the diversity and consistency of user identity information samples ($H_1, H_2, \dots, H_n$) and their average value ($\bar{H}$), as well as the matching permission parameter (ψ) that governs the compatibility and interoperability of identity information across domains.

The base parameter of user identity information value (ϑ), user behavior parameter (ν), and user behavior trajectory distribution parameter (ζ) collectively contribute to the characterization of user behavior trajectories in the HCN. These parameters capture the intrinsic properties, behavioral patterns, and distribution characteristics of user activities, enabling the system to establish a comprehensive behavioral profile for each user.

The unit coding duration ($\Delta\eta$) represents the time interval during which the user identity attribute information is encoded and processed within the HCN. This parameter ensures that the user behavior trajectory features are captured and analyzed promptly and efficiently, allowing for real-time monitoring and response to potential security threats.

The ZTB-NGHCN framework employs a tiered architecture to accommodate IoT medical devices with varying computational capabilities. Resource-constrained devices are paired with nearby edge nodes that perform computationally intensive cryptographic operations and trust calculations. A lightweight client is implemented directly on the device for more capable devices, enabling direct participation in the blockchain network while optimizing for power efficiency.

B. Blockchain and Risk Assessment-Based Access Control

The blockchain and risk assessment-based access control component leverages smart contracts to provide decentralized and intelligent access control [24], [25]. The access control component follows the blockchain and risk assessment-based access control (BR-ABAC) mechanism design and definition. It incorporates the sliding window mechanism, punishment mechanism, access subject trust value calculation, and dynamic permission threshold adjustment algorithm.

To ensure the reliability and timeliness of the calculation data in HCN, we introduce a sliding window mechanism for trust value calculation [26]. The sliding window mechanism in ZTB-NGHCN employs an adaptive algorithm that adjusts the window size based on the rate of change in user behavior and network conditions. During periods of high volatility, the window size is automatically reduced to give more weight to recent events, allowing for faster detection of potential security threats. When calculating the trust value, only the data within the sliding window is used, while the data outside the window is ignored. Each time an interaction occurs, the window moves by one unit, and the weight of the data within the window changes accordingly until it leaves the window and loses its

reference value. The following gives the decay function based on the sliding window model in HCN.

$$B(\tau) = \begin{cases} 1 - \beta & \tau = \omega \\ B(\tau - 1) + \frac{1}{\omega} & \tau < \omega \end{cases} \quad (5)$$

where ω denotes the window size in the HCN, τ denotes the distance between the historical access record and the current access record, and β denotes an arbitrarily small positive number used to adjust the decay range in the healthcare context.

The entire HCN adopts a reduction-only model to punish nodes that exhibit malicious behavior. Once the system detects an illegal operation, it will add an upper trust value limit for the user and device in the access subject, thereby limiting their subsequent access to permissions. The trust upper limit in the HCN is calculated by the punishment function, as shown in the following.

$$q = D \sin\left(\frac{\tilde{n}}{2Y} \times \pi\right) \times q' \quad (6)$$

where q' denotes the threshold or trust upper limit of the target permission of the access subject in the HCN, D denotes the punishment coefficient (a smaller coefficient indicates a greater punishment), Y denotes the number of levels included in the illegal operation (divided into five levels from low to high: lowest, lower, medium, higher, highest), and $\tilde{n}$ denotes the level corresponding to the currently illegal operation in the healthcare context.

Based on the logistic regression model, we provide a basic calculation method for the trust value of each measurement indicator of the access subject in HCN. First, we use set ν to represent the user, device, and environment in the healthcare context and convert their respective measurement indicators into corresponding trust level values, denoted by set l_ν . The following is the calculation of the basic trust value of the three in HCN.

$$h_\nu(\zeta) = \frac{1}{1 + e^{-\sum_{i \in l_\nu} b_i \times \zeta_i \times B(\tau)}} \quad (7)$$

where ζ_i denotes the trust value contained in each measurement indicator in the healthcare context, b_i denotes the influence factor (i.e., weight) of each measurement indicator, and $B(\tau)$ denotes the decay factor that causes the trust value to change as the window moves in the HCN. Thus, $h_\nu(\zeta)$ represents the comprehensive trust value of the user, device, and environment in the healthcare context, respectively.

The following is the comprehensive trust value of the access subject in the HCN calculated from the three.

$$H'(\zeta) = \sum_{\nu} d_\nu \times h_\nu(\zeta) \quad (8)$$

where d_ν denotes the weight of the user, device, and environment in the healthcare context, and the sum of the weights is 1. $H'(\zeta)$ represents the comprehensive trust value of the access subject in the HCN.

To ensure the security of the HCN, the threat behavior should also be considered when calculating the trust value. Based on the trust measurement system, the threat behavior in the healthcare context is represented by set $\Psi =$

$\psi_1, \psi_2, \dots, \psi_t$. Let t_φ denote the influence factor of each threat behavior, where $\varphi \in \Psi$ and the initial value of t_φ is 1. The factors that influence threat behavior in the HCN are calculated as follows.

$$\Theta(t) = \prod_{\varphi \in \Psi} t_\varphi. \quad (9)$$

When the HCN detects a corresponding threat behavior, it directly modifies the value of t_φ to 0, thereby modifying the trust value and preventing illegal subjects from obtaining permissions.

Therefore, the complete algorithm for calculating the comprehensive trust value of the access subject in the HCN is given below.

$$H(\zeta) = \Theta(t)H'(\zeta) = \prod_{\varphi \in \Psi} t_\varphi \times \sum_{\nu} d_\nu \times h_\nu(\zeta). \quad (10)$$

The ZTB-NGHCN framework incorporates a dynamic consent management system built on smart contracts. This system allows patients to grant, modify, or revoke consent for various data usage scenarios, including research purposes. Each consent decision is recorded on the blockchain, creating an immutable audit trail. The system also supports granular consent options, allowing patients to specify which data types can be shared and for what purposes. Researchers can then query the consent management system to ensure they only access data for which they have valid patient consent.

C. Dynamic Trust Gateway

The dynamic trust gateway component employs a consensus mechanism to achieve dynamic gateway switching and enhance network resilience. The dynamic trust gateway model follows the design in and integrates zero-trust, NGN, and HCN.

In the dynamic trust gateway model, N gateways form a private blockchain, and each gateway serves as a node. According to the blockchain consensus algorithm, the gateway that first completes the hash value calculation meets the difficulty requirement and is elected as the main gateway. The main gateway has the right to execute the access group allocation algorithm, is responsible for mining potential blocks and uploading them to the chain, and selects the access group for the terminal device access process. Usually, the gateway that completes the hash value calculation first is the gateway with the strongest computing power in the current period, and it is the optimal solution for it to serve as the main gateway during this period. Additionally, by setting different difficulty values, the replacement cycle of the main gateway can be adjusted.

The ZTB-NGHCN framework incorporates a middleware layer that facilitates seamless integration with legacy healthcare systems. This layer utilizes standardized healthcare data exchange protocols, such as HL7 FHIR, to ensure interoperability while maintaining the zero-trust security model.

The main gateway has functions similar to those of other gateways. The attack logs and other information captured by the gateway will be uploaded to the blockchain, and other nodes will synchronize the uploaded data in the private

chain. The main gateway is responsible for selecting the corresponding access group to perform identity authentication for the device during the access authentication process of the healthcare Internet of Things (IoT) devices.

The ZTB-NGHCN framework incorporates a robust key management system that leverages threshold cryptography for distributed key generation and storage. Regular key rotation is enforced through smart contracts, with a configurable rotation period based on the healthcare organization's security policies. This approach ensures that compromised keys have a limited impact on the overall system security.

The ZTB-NGHCN framework implements a distributed backup and recovery system that ensures data availability despite node failures or network partitions. Critical data is replicated across multiple geographically distributed nodes. In failures, an automatic recovery process is triggered to restore the system state from the distributed backups.

III. ZERO-TRUST NEXT-GENERATION HCN

This section describes the zero-trust next-generation HCN based on the proposed framework. It consists of three main parts: cross-domain matching of zero-trust next-generation HCN user identities, workflow of BR-ABAC mechanism, and access authentication.

A. Cross-Domain Matching of Zero-Trust Next-Generation HCN User Identities

The network functional area division criteria determine the storage and carrying capacity of the network host for user identity information in HCN. The functional area division conditions directly affect the coding of the user identity information original ID code by the host element. If the HCN based on the zero-trust security architecture is considered as a complete data storage space, its storage capacity for information samples can be expressed as follows.

$$\Gamma_\kappa = \left\| \frac{\Lambda_\kappa}{\Gamma_\kappa} \right\|. \quad (11)$$

where κ denotes the network security rating result in the HCN, Λ_κ denotes the network rating vector under coefficient κ , and Γ_κ denotes the storage characteristics of user identity information in the healthcare context.

Let $\bar{\Sigma}$ denote the network area limit parameter in the HCN and θ_κ denote the user identity information running storage parameter under coefficient κ . Then, the network functional area division conditions based on the zero-trust security architecture in HCN can be expressed as follows.

$$\Phi = \frac{|\Omega \cdot \Gamma_\kappa|}{(\kappa^2 - 1) \times \bar{\Sigma} \cdot \theta_\kappa}, \kappa \neq 0. \quad (12)$$

In HCN, the network functional area division is crucial in ensuring the secure and efficient storage and processing of user identity information [27]. By dividing the network into distinct functional areas based on the zero-trust security architecture, the system can allocate resources and apply security measures in a targeted and effective manner.

The storage capacity for information samples, denoted by Γ_κ , is a key factor in determining the network's ability to

handle user identity information in healthcare. This capacity is influenced by the network security rating result (κ), the network rating vector (Λ_κ), and the storage characteristics of user identity information (Γ_κ) specific to the healthcare domain.

The network area limit parameter ($\bar{\Sigma}$) and the user identity information running storage parameter (θ_κ) are essential components in defining the functional area division conditions.

By incorporating these healthcare-specific parameters into the network functional area division process, the zero-trust security architecture can be tailored to meet the specific needs of HCN. This customization ensures that user identity information is stored and processed securely, efficiently, and competently while facilitating cross-domain authentication and access control.

Eq. (12) represents the network functional area division conditions based on the zero-trust security architecture in HCN. It takes into account the storage capacity (Γ_κ), the network security rating result (κ), the network area limit parameter ($\bar{\Sigma}$), and the user identity information running storage parameter (θ_κ). Combining these factors allows the system to determine the optimal functional area division that balances security, performance, and resource utilization.

The network functional area division in HCN is critical to implementing a robust and effective zero-trust security architecture. By carefully considering the unique characteristics and requirements of the healthcare domain, the system can ensure the secure and efficient management of user identity information, enabling seamless cross-domain authentication and access control while maintaining the highest data privacy and protection standards.

In the zero-trust security architecture of HCN, the matching call chain code is responsible for linking multiple independent cross-domain nodes together so that the network host can complete the selection and matching of user identity information according to the predetermined data arrangement pattern. Let $\eta_1, \eta_2, \dots, \eta_n$ denote n independent cross-domain nodes in the HCN, and the definition condition is as follows.

$$\eta_1 \neq \eta_y \neq \dots \neq \eta_n. \quad (13)$$

where y denotes the node call parameter in the HCN, its solving criterion satisfies the following.

$$y = \frac{g \cdot \varphi}{C}. \quad (14)$$

where g denotes the chain connection coefficient in the HCN, φ denotes the user identity information placeholder parameter, and C denotes the cross-domain transmission standard value of the information sample in the healthcare context.

Combining Eqs. (13) and (14), the network user identity cross-domain matching result in the HCN, i.e., the matching call chain code, is given by the following.

$$\Psi = \Phi \times \frac{\eta_1 \times \eta_2 \times \dots \times \eta_n}{y \cdot \|\varphi\|^2}. \quad (15)$$

where φ denotes the processing weight of network user identity information cross-domain matching in the HCN.

The matching call chain code-solving process is a critical component in the zero-trust security architecture of HCN.

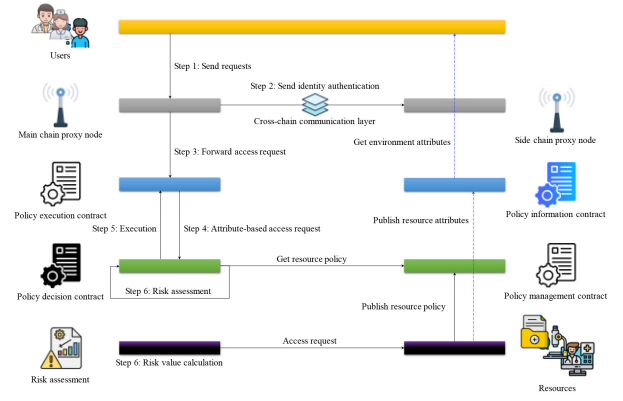


Fig. 2. Workflow of BR-ABAC mechanism in zero-trust NGN-enabled HCN.

It enables the efficient and secure matching of user identity information across different domains, facilitating seamless authentication and access control.

In HCN, the independent cross-domain nodes ($\eta_1, \eta_2, \dots, \eta_n$) represent the various entities, systems, or departments involved in the user identity management process. These nodes may include healthcare providers, insurance companies, government agencies, and other relevant stakeholders.

The node call parameter (y) plays a crucial role in determining the sequence and priority of the cross-domain nodes during the matching process. It considers the chain connection coefficient (g), which represents the strength and reliability of the links between the nodes, and the user identity information placeholder parameter (φ), which ensures the integrity and consistency of the user identity information across different domains.

The cross-domain transmission standard value (C) sets the benchmark for transmitting and exchanging user identity information in the HCN. It ensures that the matching call chain code-solving process adheres to the necessary security and privacy standards specific to the healthcare domain.

B. Workflow of BR-ABAC Mechanism in Zero-Trust NGN-Enabled HCN

The workflow of the BR-ABAC mechanism in zero-trust NGN-enabled HCN follows the specific workflow of the BR-ABAC mechanism, as shown in Fig. 2. It integrates zero-trust principles, NGN technologies, and HCN requirements to ensure secure and efficient access control.

The detailed workflow is as follows:

Step 1: The user requests access to resources within the zero-trust NGN-enabled HCN. The request parameters include the user's unique identifier (UID), identity information signature (signInfo), resource identifier (RID), and the operation to be executed (AAttr).

Step 2: Upon receiving the request, the main chain proxy node, which acts as a trusted authority in the zero-trust architecture, parses the user's UID to obtain the sub-chain ID to which the user belongs. It then sends an identity authentication request to the corresponding sub-chain through the cross-chain communication layer, which leverages NGN's

high-speed and low-latency capabilities. The request parameters include signInfo and UID.

Step 3: If the identity authentication is successful, the main chain proxy node forwards the access request to the policy execution contract deployed on the blockchain. This triggers the execution of the policy execution contract, which is designed to enforce access control policies in the zero-trust NGN-enabled HCN. The request parameters include the user's UID, RID, and AAttr.

Step 4: The policy execution contract converts the request into an attribute-based access request (AAR) that adheres to the BR-ABAC mechanism. It then invokes the policy decision contract for decision-making and receives the decision result. The policy decision contract evaluates the request based on the user's attributes, resource attributes, and environmental conditions, following the principles of zero-trust security.

Step 5: After receiving the attribute-based access request, the policy decision contract first invokes the policy information contract to obtain the access subject attribute information based on the user's UID. It also retrieves the resource attribute information based on the RID and the environment attribute information specific to the HCN. Subsequently, it invokes the policy management contract to obtain the resource access policy information through the RID. Finally, it decides based on the attribute set information and the resource access policy information. The decision result is then returned to the policy execution contract.

Step 6: The policy execution contract evaluates the result of the returned decision. If the decision grants permission, the AAR is sent to the risk assessment layer. At this layer, risk features are extracted and input into the access risk assessment model for real-time risk evaluation. The risk assessment model leverages advanced technologies, such as machine learning and big data analytics, to accurately determine the risk level associated with the access request in the context of the zero-trust NGN-enabled HCN.

Step 7: The access risk assessment model evaluates the access risk of the user's current access request and calculates the risk value. Predefined risk thresholds and policies determine whether to allow the user to perform the requested operation on the resource. Access is immediately denied if the risk level exceeds a critical threshold and the user's access rights are revoked. This dynamic risk assessment and access control mechanism ensures that the zero-trust principles are enforced throughout the access lifecycle in the HCN.

The BR-ABAC mechanism workflow provides a robust and adaptive access control solution by integrating zero-trust principles, NGN technologies, and HCN requirements. It leverages NGN's high-speed and low-latency capabilities to enable fast and efficient authentication and authorization processes. Blockchain technology ensures access control policies and decisions' integrity, transparency, and immutability.

C. Access Authentication

The access authentication process follows the access and authentication processes, integrating zero-trust, NGN, and HCN.

In the context of zero-trust NGN-enabled HCN, the access process for legitimate access terminals is managed by different gateways according to their configuration and region. Each legitimate access request to the healthcare IoT business system forms a block in the blockchain, ensuring the integrity and traceability of access records.

In addition to the information required to form a chain storage structure, the block contains crucial metadata specific to the zero-trust NGN-enabled HCN. This metadata includes the timestamp when the node accesses the business system, the DID, business status, public key and digital certificate, permission level information, and the main gateway information corresponding to the current access request.

The integration of zero-trust principles in the access process ensures that every access request, regardless of origin, is treated as untrusted by default. Each request undergoes strict authentication and authorization procedures before being granted access to the healthcare IoT business system.

The ZTB-NGHCN framework includes an 'emergency access protocol' to handle emergencies. The EAP allows authorized healthcare professionals to elevate their access privileges in critical situations temporarily. A combination of biometric authentication and a secondary verification from an on-call administrator triggers this process. All emergency access events are logged with high-priority flags in the blockchain for immediate post-event audit and justification review.

Request Stage: When a healthcare IoT device α requires access authentication, it initiates the process by sending an authentication request to its associated regional trust gateway TG_α . The request includes a digital signature Σ of the device's unique identifier DID_α and the device's public key PK_α . The DID_α is a distinctive identifier that encapsulates information about the device's healthcare-related attributes and functionalities. Upon receiving the request, the regional trust gateway TG_α verifies the device's signature using the provided public key PK_α . If the verification is successful, TG_α encapsulates the authentication request and forwards it to the main gateway MG designated for the current time interval ΔT .

Verification Stage: Upon receiving the authentication request from the regional trust gateway TG_α , the main gateway MG selects a group of qualified nodes within its region to form an authentication group $\Gamma = \Theta_1, \Theta_2, \dots, \Theta_n$. The selection criteria for the authentication group nodes consider factors such as their healthcare-related roles, processing capabilities, and trust levels. MG then transmits the authentication request to the nodes in group Γ using a secure multicast communication protocol enabled by NGN technologies. The nodes in Γ collaboratively execute a distributed consensus algorithm, such as the practical Byzantine fault tolerance algorithm, to agree on the device's authenticity α . During this process, the nodes utilize the device's public key certificate and a voting-based consensus mechanism to validate the device's identity and generate a new block containing the authentication information.

The ZTB-NGHCN framework employs a hierarchical consensus mechanism to ensure scalability in large-scale

healthcare networks. This approach divides the network into clusters, each running a fast BFT consensus algorithm. Inter-cluster consensus is achieved through a delegated proof-of-stake mechanism, allowing the system to maintain high throughput and low latency even as the network grows.

As part of the authentication process, a (k, n) threshold secret sharing scheme is employed to distribute and reconstruct the secret information Ω (i.e., the digital certificate issued by the trusted authority to the healthcare IoT device). In this scheme, k represents the minimum number of nodes required to reconstruct the secret, and n represents the total number of shares the secret is divided into. For the HCN, we set $k = n$ to ensure high security. The secret sharing scheme guarantees that only when k or more nodes collaborate can they successfully reconstruct the secret information Ω , while any number of nodes fewer than k cannot reveal any information about Ω .

In the pre-distribution stage, the main gateway MG selects n nodes and assigns them unique coefficients $\beta_1, \beta_2, \dots, \beta_n \in \mathbb{Z}_p^*$. Each node Θ_i receives a share of the secret calculated using the following.

$$\Lambda(x_i) = \Omega + \beta_1 x_i^1 + \beta_2 x_i^2 + \dots + \beta_{(n-1)} x_i^{(n-1)}. \quad (16)$$

where $\Lambda(x_i)$ represents the share allocated to node Θ_i .

Each node in the authentication group Γ can independently verify the reconstructed secret information Ω using the public key of the trusted authority responsible for issuing digital certificates to healthcare IoT devices.

Confirmation Stage: Once the authentication process is completed, the nodes in the authentication group Γ send their individual authentication results to the main gateway MG . Based on the received results, MG makes a final decision on the authenticity of the healthcare IoT device α and generates a new block containing the authentication information. This block is then appended to the blockchain, creating an immutable record of the authentication event. Finally, MG returns the authentication decision to the device α , concluding the authentication process.

While the ZTB-NGHCN framework enhances security, it is crucial to consider potential privacy implications. The continuous monitoring and data collection required for zero-trust implementation must be balanced with patient privacy rights and regulatory compliance, such as Health Insurance Portability and Accountability Act (HIPAA) in the United States or General Data Protection Regulation (GDPR) in Europe. The ZTB-NGHCN framework has been designed with regulatory compliance in mind. It incorporates data minimization, purpose limitation, and user consent management to align with HIPAA and GDPR requirements. Additionally, the blockchain's immutable audit trail facilitates compliance reporting and breach notification processes.

To facilitate compliant cross-border data sharing, the ZTB-NGHCN framework incorporates a dynamic policy engine that automatically adjusts data access and sharing rules based on geographical location and applicable regulations. This engine integrates with a regularly updated international healthcare data protection law database, ensuring that data transfers always comply with relevant jurisdictions.

TABLE I
TRUST CALCULATION INDICATORS FOR HCN

Indicator category	Indicator name	Description
User indicators	U1	User role and privileges
	U2	User behavior patterns
	U3	User authentication history
Device indicators	D1	Device type and capabilities
	D2	Device security posture
	D3	Device location and network access
Network indicators	N1	Network throughput and latency
	N2	Network congestion and error rates
	N3	Network security events and anomalies

IV. SIMULATION AND RESULTS ANALYSIS

A. Simulation Settings

We conducted extensive simulations to demonstrate the superior performance of the proposed ZTB-NGHCN framework in zero-trust blockchain-enabled secure next-generation HCN. We compared the results with six baseline methods: Zt&t [28], LBFT [29], BZTIoT [30], VSSB-Raft [31], BAC-LDS [32], and ZT-IIoT [33]. Zt&t introduces a secure service session management mechanism that leverages blockchain technology. Alongside, LBFT proposes a novel asynchronous committee-based blockchain storage strategy termed lightweight BFT, which optimizes system performance and is versatile across various scenarios. BZTIoT delivers an innovative approach to blockchain-facilitated information sharing within zero-trust IoT environments. Additionally, VSSB-Raft provides a robust and efficient consensus algorithm known as the verifiable secret sharing Byzantine fault tolerance Raft consensus algorithm. Furthermore, BAC-LDS is a blockchain-based access control scheme that facilitates lightweight data sharing among diverse organizations. Finally, ZT-IIoT offers a comprehensive implementation framework for zero-trust verification of IIoT wireless transmission nodes, enhancing security in digitally interconnected systems. The simulations were performed using MATLAB, and the experimental environment was configured with an Intel Core i9-10900K CPU @3.70GHz, 64GB memory, Windows 10 system, and MATLAB R2020a.

We evaluated the performance of the ZTB-NGHCN framework through various simulations, considering the unique aspects of zero-trust, NGN, and HCN. The trust calculation indicators and experimental parameter settings used in the simulations are shown in Tables I and II, respectively.

The simulations focused on three key aspects: the impact of measurement indicators on trust value calculation, the impact of the sliding window on trust value, and the impact of the punishment mechanism on trust value. Additionally, we evaluated the attack detection accuracy, access control granularity, and

TABLE II
EXPERIMENTAL PARAMETER SETTINGS FOR HCN

Parameter	Value
Number of users	1000
Number of attributes	10
User behavior trajectory feature size	50
Number of access control policies	100
Number of attributes in policies	5
Number of risk assessment rules	20
Number of gateways	10
Number of consensus rounds	3
Simulation time	1000 seconds

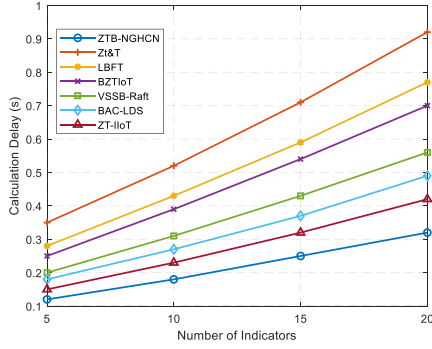


Fig. 3. Impact of measurement indicators on calculation delay.

network resilience of the ZTB-NGHCN framework compared to the six baseline methods.

In Table II, the 1000 users were generated synthetically to represent a diverse healthcare network. They were randomly assigned roles (e.g., doctors, nurses, administrators) and attributes (e.g., department, access level) based on typical distributions in healthcare settings. This ensured a realistic representation of a large-scale HCN.

The key mathematical parameters used in our simulations include the trust decay factor as 0.9, the punishment coefficient as 0.8, the risk assessment threshold as 0.7, and the consensus threshold as 0.75. These parameters were chosen based on preliminary experiments and literature recommendations to balance security and performance.

B. Results and Analysis

1) *Impact of Measurement Indicators on Trust Value Calculation*: We analyzed the impact of the number of measurement indicators on the calculation delay of the trust value. The results are shown in Fig. 3. As the number of measurement indicators increases, the calculation delay for all methods increases. However, the ZTB-NGHCN framework consistently outperforms the baseline methods, demonstrating lower calculation delays. This is attributed to the efficient trust calculation algorithm and the distributed processing capabilities of the blockchain infrastructure.

2) *Impact of Sliding Window on Trust Value*: We evaluated the impact of the sliding window size on the trust

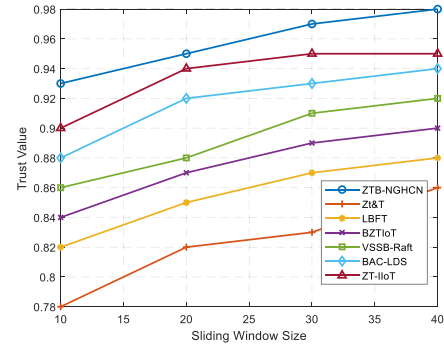


Fig. 4. Impact of sliding window on trust value.

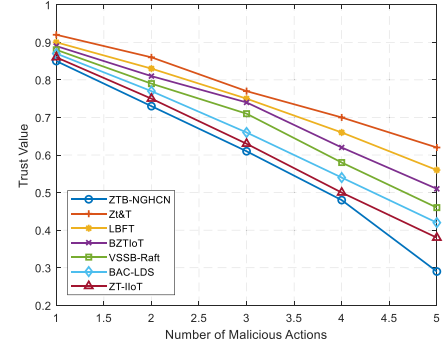


Fig. 5. Impact of punishment mechanism on trust value.

value calculation. The results are presented in Fig. 4. The ZTB-NGHCN framework achieves higher trust values than the baseline methods for all sliding window sizes. As the sliding window size increases, the trust value for all methods improves, indicating that considering a larger history of user behavior and network events enhances the accuracy of trust assessment. The superior performance of ZTB-NGHCN is due to its comprehensive trust measurement system, which considers various user, device, and network indicators.

3) *Impact of Punishment Mechanism on Trust Value*: We investigated the impact of the punishment mechanism on the trust value of malicious users. The results are shown in Fig. 5. The ZTB-NGHCN framework exhibits a more rapid decrease in trust value as malicious actions increase. This demonstrates the effectiveness of the punishment mechanism in penalizing malicious users and reducing their trust scores.

4) *Attack Detection Accuracy*: We evaluated the attack detection accuracy of the ZTB-NGHCN framework and the baseline methods by measuring the true positive rate (TPR) and false positive rate (FPR). The results are presented in Fig. 6. The ZTB-NGHCN framework achieves the highest TPR and the lowest FPR among all methods, demonstrating its superior attack detection accuracy. The high TPR indicates that ZTB-NGHCN successfully identifies a large proportion of actual attacks, while the low FPR suggests that it minimizes the misclassification of non-attacks as attacks.

5) *Access Control Granularity*: We assessed the access control granularity of the ZTB-NGHCN framework and the baseline methods by measuring the success rate of authorized access attempts and the rejection rate of unauthorized access attempts. The results are shown in Fig. 7. The ZTB-NGHCN

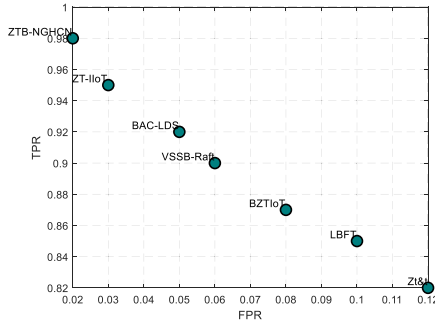


Fig. 6. Attack detection accuracy comparison.

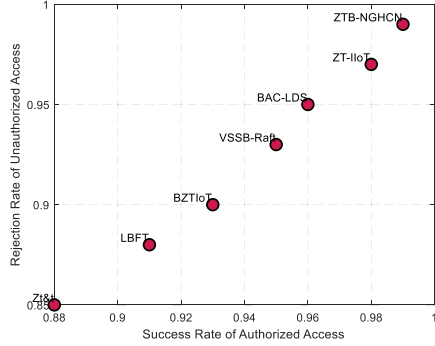


Fig. 7. Access control granularity comparison.

framework achieves the highest success rate of authorized access attempts and the highest rejection rate of unauthorized access attempts, indicating its superior access control granularity.

6) *Network Resilience*: We evaluated the network resilience of the ZTB-NGHCN framework and the baseline methods by measuring the average transaction throughput and latency under different node failure and attack scenarios. The results are presented in Figs. 8 and 9. The ZTB-NGHCN framework maintains higher transaction throughput and lower latency than the baseline methods under various failure and attack scenarios. Even in the presence of node failures or malicious nodes, ZTB-NGHCN demonstrates minimal degradation in performance, indicating its strong network resilience. The resilience of ZTB-NGHCN is achieved through its dynamic trust gateway component, which employs a robust consensus mechanism and ensures the continued operation of the blockchain network even under adverse conditions.

7) *Scalability Analysis*: We evaluated the scalability of the ZTB-NGHCN framework by measuring the transaction throughput and latency as the network nodes increase. The results are shown in Figs. 10 and 11. The ZTB-NGHCN framework demonstrates superior scalability compared to the baseline methods. As the number of nodes increases, ZTB-NGHCN achieves higher transaction throughput and maintains lower latency, indicating that the framework can efficiently handle a growing number of participants in the HCN without significant performance degradation. The scalability of ZTB-NGHCN is attributed to its optimized blockchain architecture, efficient consensus mechanism, and distributed processing capabilities.

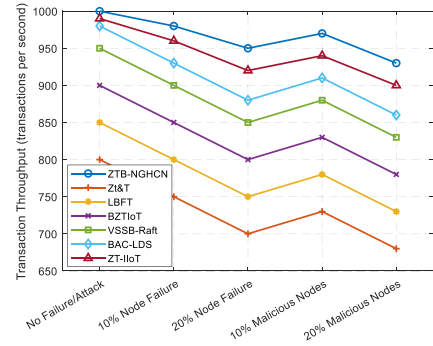


Fig. 8. Average transaction throughput under different failure and attack scenarios.

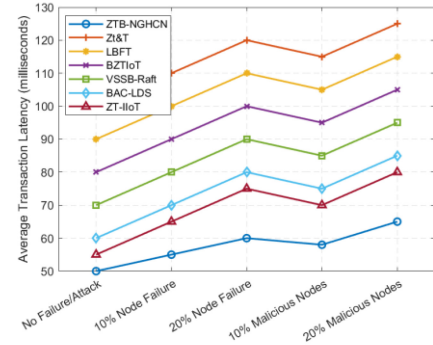


Fig. 9. Average transaction latency under different failure and attack scenarios.

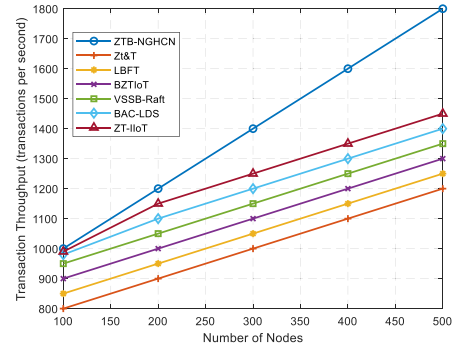


Fig. 10. Transaction throughput vs. number of nodes.

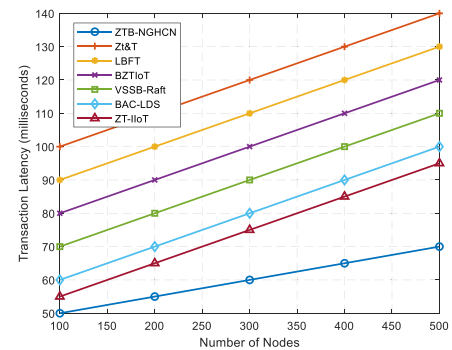


Fig. 11. Transaction latency vs. number of nodes.

8) *Energy Consumption Analysis*: We analyzed the energy consumption of the ZTB-NGHCN framework and the baseline methods by measuring the average energy consumed per

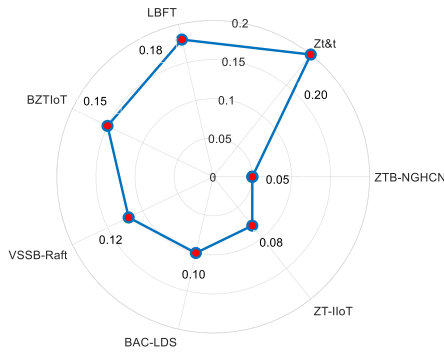


Fig. 12. Energy consumption per transaction.

TABLE III
IMPACT OF ZTB-NGHCN ON HEALTHCARE PROFESSIONALS' WORKFLOW

Metric	Before	After	Change
Average time to access patient records	45 seconds	12 seconds	-73.30%
Authentication-related disruptions/day	8.5	2.3	-72.90%
Reported security incidents/month	3.2	0.5	-84.40%
Time spent on security procedures/day	28 minutes	11 minutes	-60.70%

transaction. The results are presented in Fig. 12. The ZTB-NGHCN framework exhibits the lowest energy consumption per transaction among all methods, indicating that ZTB-NGHCN is energy-efficient and can process transactions with minimal energy overhead. The low energy consumption of ZTB-NGHCN is achieved through its optimized blockchain consensus mechanism, efficient data structures, and intelligent resource allocation strategies. By consuming less energy per transaction, ZTB-NGHCN contributes to the sustainability and eco-friendliness of the HCN.

To assess the impact of the ZTB-NGHCN framework on healthcare professionals' workflow and user experience, we conducted a mixed-methods study involving 100 healthcare professionals from various specialties over a 3-month period. The study included quantitative metrics on system usage and qualitative feedback through surveys and interviews. Table III presents the key findings from our quantitative analysis.

The results indicate significant improvements in workflow efficiency and security. The dramatic reduction in time to access patient records (73.3% decrease) can be attributed to the continuous authentication mechanism, eliminating the need for repeated logins. This is further reflected in the 72.9% decrease in daily authentication-related disruptions. The substantial decrease in reported security incidents (84.4% reduction) suggests that the zero-trust model effectively prevents unauthorized access attempts. Additionally, the time spent on security procedures decreased by 60.7%, indicating that the automation and intelligence built into the ZTB-NGHCN framework reduce the manual security burden on healthcare professionals.

The simulation results demonstrate the effectiveness of the ZTB-NGHCN framework in addressing key security

challenges in next-generation HCN. By integrating zero-trust principles with blockchain technology, our framework achieves superior performance in trust calculation, attack detection, access control, and network resilience. These improvements have significant implications for the healthcare industry, potentially enhancing patient data security, streamlining healthcare operations, and enabling more secure telemedicine and remote monitoring applications. Future work could explore the framework's applicability to other critical infrastructure sectors and investigate its performance in real-world healthcare environments.

V. CONCLUSION

In this paper, the proposed ZTB-NGHCN framework integrated zero-trust principles and blockchain technology to provide a decentralized, secure, and intelligent solution for healthcare communication in NGNs. We conducted extensive simulations and comparative analyses with baselines to demonstrate the superior performance of the ZTB-NGHCN framework in various aspects. However, the proposed framework was evaluated through simulations, and real-world implementation and testing would provide further insights into its performance and practicality. Furthermore, exploring the incorporation of privacy-preserving techniques, could strengthen the privacy protection capabilities of the framework.

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